EE1007
Introduction to Computer Science Laboratory

Lab 5
Control statement II (repeat control)

Andy, Yu-Guang Chen
Assistant Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw

Lab 5.1
P113017 Lab 5.1: Calculator (30%)

- Test target: switch(…){…}
- You are asked to write a program to create a calculator. The program will ask the user to enter a mathematical operator in the following table. Next, the user needs to enter two numbers which will be calculated by the mathematical operator. Then, output the result.
- The program will continue running the above steps until the user enters letter e.

Lab 5.1
P113017 Lab 5.1: Calculator (30%)

- Input
- Input the operator (+, -, *, ^, and e).
- Input two integers which range in [0, 2^31-1].

- Output
- Print the calculation result.
- Error message.

Lab 5.1
P113017 Lab 5.1: Calculator (30%)

- thought
- Find what the program keeps repeating. Observe that entering the operator and checking if the input is correct are being repeated.
  - Use while loop to implement the program.
  - checking if the input is correct → Use switch to check the correctness.
- Two integers which range in [0, 2^31-1] do multiplication. The calculation result will exceed range [0, 2^31-1].
  - Transfer the int type into long long int type before multiplication. → Use static_cast<long long int>(…).

Lab 5.1
P113017 Lab 5.1: Calculator (30%)

- Declare the variables

```
char m_operator;//mathematical operator
int num_a, num_b;//two number which will be calculate by the mathematical operator
long long int calculation_result;
```


Lab 5.1
P113017 Lab 5.1: Calculator (30%)

- In a while loop, a switch checks the correctness of the input.

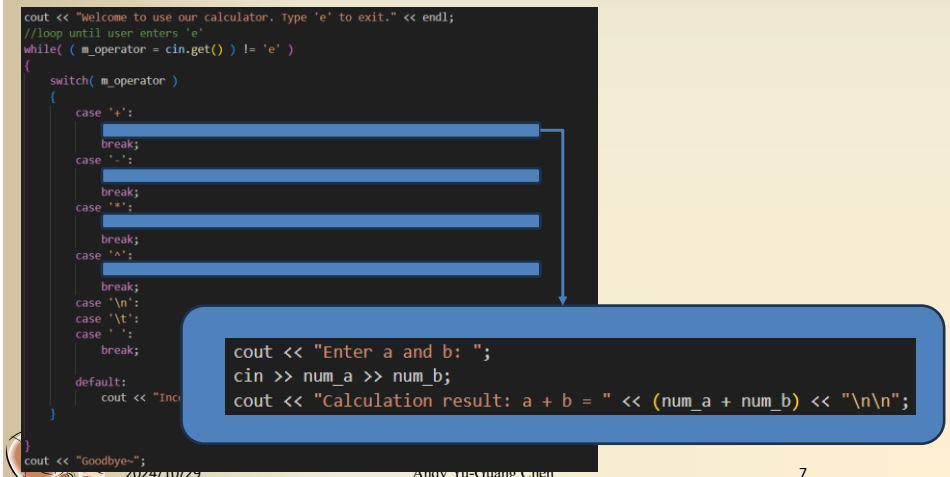
```
cout << "Welcome to use our calculator. Type 'e' to exit." << endl;
//loop until user enters 'e'
while( ( m_operator = cin.get() ) != 'e' )
{
    switch( m_operator )
    {
        case '+':
            [redacted]
            break;
        case '-':
            [redacted]
            break;
        case '*':
            [redacted]
            break;
        case '^':
            [redacted]
            break;
        case '\n':
        case '\t':
        case ' ':
            break;

        default:
            cout << "Incorrect mathematical operator entered. Enter a new one." << endl;
    }

}
cout << "Goodbye~";
```

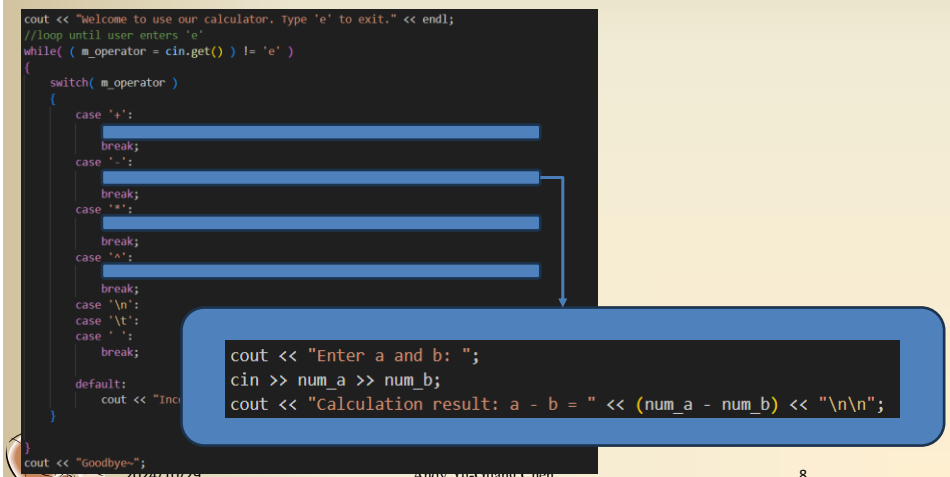
Lab 5.1
P113017 Lab 5.1: Calculator (30%)

◆ If the input is correct, input two integers and calculate the result.



Lab 5.1
P113017 Lab 5.1: Calculator (30%)

◆ If the input is correct, input two integers and calculate the result.



Lab 5.1
P113017 Lab 5.1: Calculator (30%)

◆ If the input is correct, input two integers and calculate the result.

```
cout << "Welcome to use our calculator. Type 'e' to exit." << endl;
//loop until user enters 'e'
while( ( m_operator = cin.get() ) != 'e' )
{
    switch( m_operator )
    {
        case '+':
            [redacted]
            break;
        case '-':
            [redacted]
            break;
        case '*':
            [redacted]
            break;
        case '^':
            [redacted]
            break;
        case '\n':
        case '\t':
        case ' ':
            break;
        default:
            cout << "Inc
    }
}
cout << "Goodbye~";
```

```
cout << "Enter a and b: ";
cin >> num_a >> num_b;
cout << "Calculation result: a * b = " << (static_cast<long long int>(num_a) * num_b) << "\n\n";
```


Lab 5.1
P113017 Lab 5.1: Calculator (30%)

◆ If the input is correct, input two integers and calculate the result.

```
cout << "Welcome to use our calculator. Type 'e' to exit." << endl;
//loop until user enters 'e'
while( ( m_operator = cin.get() ) != 'e' )
{
    switch( m_operator )
    {
        case '+':
            [redacted]
            break;
        case '-':
            [redacted]
            break;
        case '*':
            [redacted]
            break;
        case '^':
            [redacted]
            break;
        case '\n':
        case '\t':
        case ' ':
            break;
        default:
            cout << "Inc
    }
}
cout << "Goodbye~";
```

```
cout << "Enter a and b: ";
cin >> num_a >> num_b;
//calculate power
calculation_result = 1;//initialize the value to 1
for ( int i = 0 ; i < num_b ; i++ )
    calculation_result = calculation_result * num_a;
cout << "Calculation result: a ^ b = " << calculation_result << "\n\n";
```

Lab 5.2
P113018 Lab 5.2: Combination (30%)

- Test target: loop(do while{…}; and for)
- You are asked to write a program to help Joe find all the possible ways that he can finish the exact number of cookies his mom gave him.



Lab 5.2
P113018 Lab 5.2: Combination (30%)

Input

- Total number of the cookies
- Three different quantities of cookies Joe can eat per day

Output

- Print the error message “Wrong quantities. Enter three quantities again.”
- Print all the possible ways to finish the cookies according to Rule 1 and 2.



Lab 5.2
P113018 Lab 5.2: Combination (30%)

- Thought - 1
- **How to do fool-proof approach?**
- Use do while{}; to ask the user input the quantities until the input is within the boundary.



Lab 5.2
P113018 Lab 5.2: Combination (30%)

- Thought - 2
- **How to find the possibilities and list them according to Rule 1 and 2 ?**
- To meet the Rule 1 and 2, we need to sort the quantities in ascending order.
- Then, the most important thing is that we need to find how many days are needed to eat each quantity separately.
  - Take the example in the OJ problem
  - Eat 2, 3, and 6 cookies for n1, n2, and n3 days respectively
  - Find (n1, n2, n3) such that Joe can finish the exact number of cookies



Lab 5.2
P113018 Lab 5.2: Combination (30%)

- Thought - 2
- In Rule 1, the case with fewer days of eating the least quantity of cookies (in this case, 2) should be listed first.
  - Take the example in the OJ problem
  - Eat 2, 3, and 6 cookies for n1, n2, and n3 days respectively
  - Try n1 from 0 to the maximum(n1*2 <= total_cookies)
- If multiple cases have the same number of days with the least quantity, the case that uses fewer days overall is listed first.
  - Try n2 from 0 to the maximum(n1*2+ n2*3 <= total_cookies)
  - If the number of the rest cookies can be divisible by 6, this is one of the solution.

Lab 5.2
P113018 Lab 5.2: Combination (30%)

- Thought - 3
- **How to print each case according to the Rule 2?**
- Rule 2 :
  - (n1, n2, n3) = (3, 2, 0)
  - [2, 2, 2, 3, 3] → Print 2 for 3 times, 3 for 2 times, 6 for 0 times

- First, print a "[" symbol.
- Next, print 2 for n1 times. Each time after printing 2, check whether it's the last one to print. If it's the last one to print, then print "]". Otherwise, print ",".

Lab 5.2
P113018 Lab 5.2: Combination (30%)

- Input and do fool-proof

```
long long int total_cookies;
long long int quant_1, quant_2, quant_3;//three different quantities of the cookies eaten per day

cout << "Enter the total number of the cookies: ";
cin >> total_cookies;
do{
    cout << "Enter 3 different quantities: ";
    cin >> quant_1 >> quant_2 >> quant_3;

    //fool-proof
    if ( quant_1 == quant_2 || quant_2 == quant_3 || quant_1 == quant_3 ) //not different
        cout << "Wrong quantities. Enter three quantities again.\n";
    else if ( quant_1 < 1 || quant_1 > 10 ) //exceed the range [1, 10]
        cout << "Wrong quantities. Enter three quantities again.\n";
    else if ( quant_2 < 1 || quant_2 > 10 )
        cout << "Wrong quantities. Enter three quantities again.\n";
    else if ( quant_3 < 1 || quant_3 > 10 )
        cout << "Wrong quantities. Enter three quantities again.\n";
    else //meet all the rules
        break;
}while( 1 );
```


Lab 5.2
P113018 Lab 5.2: Combination (30%)

- Sort the quantities in ascending order: quant_1 < quant_2 < quant_3

```
//sort the quantities in ascending order: quant_1 < quant_2 < quant_3
long long int temp;
//1. the biggest one is swapped to the last position
if ( quant_1 > quant_2 ){//swap 1 and 2
    temp = quant_1;
    quant_1 = quant_2;
    quant_2 = temp;
}
if ( quant_2 > quant_3 ){//swap 2 and 3
    temp = quant_2;
    quant_2 = quant_3;
    quant_3 = temp;
}
//2. swap the smallest one into the first position
if ( quant_1 > quant_2 ){//swap 1 and 2
    temp = quant_1;
    quant_1 = quant_2;
    quant_2 = temp;
}
```

Lab 5.2
P113018 Lab 5.2: Combination (30%)

- Print all the possible ways

```
//print all the possibilities
bool possible = false;
cout << "\nAll possibilities:\n";
for ( int day_1 = 0 ; day_1*quant_1 <= total_cookies ; day_1++ )
{
    for ( int day_2 = 0 ; day_1*quant_1+day_2*quant_2 <= total_cookies ; day_2++ )
    {
        //if the remaining cookies can be divided by quant_3 with no remainder
        //this case is one of the solution
        int remaining_cookies = total_cookies - (day_1*quant_1 + day_2*quant_2);
        if ( remaining_cookies % quant_3 == 0 )
        {
            possible = true;
            int day_3 = remaining_cookies / quant_3;
            //print this case
            cout << "[";
            for ( int i = 1 ; i <= day_1 ; i++ )
                cout << quant_1 << ((i*quant_1 != total_cookies)?", ":"]\n");
            for ( int i = 1 ; i <= day_2 ; i++ )
                cout << quant_2 << ((day_1*quant_1+i*quant_2 != total_cookies)?", ":"]\n");
            for ( int i = 1 ; i <= day_3 ; i++ )
                cout << quant_3 << ((day_1*quant_1+day_2*quant_2+i*quant_3 != total_cookies)?", ":"]\n");
        }//end if

    }//end for day_2
}//end for day_1

if ( !possible )
    cout << "There is no way to finish the cookies.";
```


Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Test target: Debugging
- The problem provides the complete source code for you, but there are several errors in the C++ code that you need to debug.
- The program will ask the user to enter today's date and how many bottles of the juice. Then, the user will enter a 7-digit number that each digit represents how many bottles to drink on each day of the week.
- In the end, the program will output the day on which all of the juice will be finished. And, print which day of the week on that date.

Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Input
  - Enter the date and which day of the week today
  - Enter how many bottles of the juice
  - Enter 7-digit number(Each digit represents how many bottles to drink on each day of the week)
- Output
  - Error message
  - Output the day on which all the juice will be finished and indicate which day of the week corresponds to that date

Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Thought - 1
- Syntax error
  - Add “#” before include.

main.cpp

```
1  #include <iostream>
2  include <cmath>
3
4  using namespace std;
5
6  int main()
7  {
8      int year, month, date;//the date today
9      int week;//which day of the week today, 1~7 => Mon.~Sun.
10     int bottle_juice;//How many bottles of the juice
```

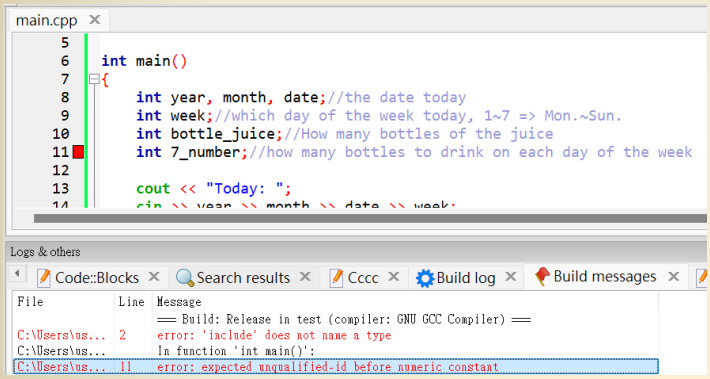
Logs & others

Code::Blocks | Search results | Cccc | Build log | Build messages

File	Line	Message
		== Build: Release in test (compiler: GNU GCC Compiler) ==
C:\Users\us...	2	error: 'include' does not name a type

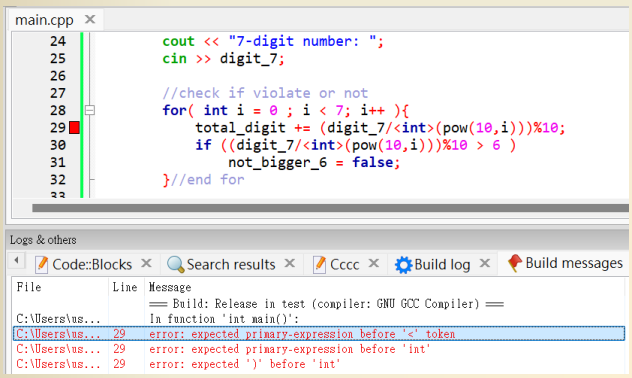
Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Syntax error
  - Naming rule: A variable name cannot start with a number.
  - Change all the 7_number to another name (ex: digit_7).



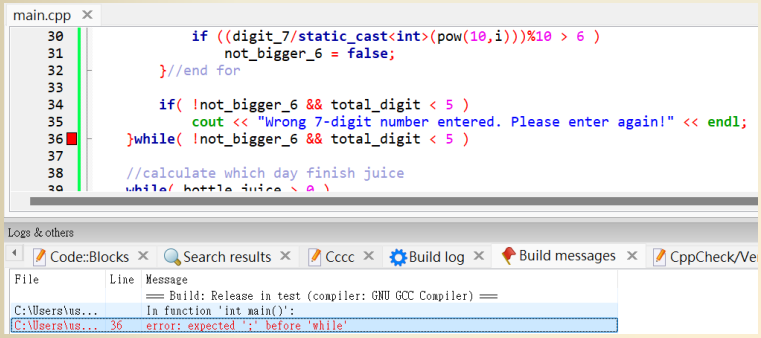
Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Syntax error
  - Add static_cast before <int>.



Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Syntax error
  - Add ";" after while(…).



Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Thought-2
- Logic error
- Screen after running

```
Today: 2024 12 31 2
How many bottles of the juice: 44
7-digit number: 5975110
Finish all the juice on 2025/01/09 (Thu.). (Fri.). (Sat.). (Sun.).
```

  - Fool-proof approach fails (&& → | |)

```
    if( !not_bigger_6 && total_digit < 5 )
        cout << "Wrong 7-digit number entered. Please enter again!" << endl;
}while( !not_bigger_6 && total_digit < 5 );
```

  - Fail to print the week
    (Need break; in each case)

```
switch( week )
{
case 1:
    cout << " (Mon.).";
case 2:
    cout << " (Tue.).";
case 3:
    cout << " (Wed.).";
case 4:
    cout << " (Thu.).";
case 5:
    cout << " (Fri.).";
case 6:
    cout << " (Sat.).";
case 7:
    cout << " (Sun.).";
    break;
}//end switch
```

Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Logic error
- Screen after running
  - Lack a empty line.

```
Today: 2024 09 03 2
How many bottles of the juice: 47
7-digit number: 0512002
Finish all the juice on 2024/10/04 (Fri.).
```

  - Add a '\n' before Finish.

```
//output the result
cout << "Finish all the juice on ";
```


Lab 5.3
P113019 Lab 5.3: Debugging (30%)

- Logic error
- Screen after running
  - Date is one day more. 2024/10/03(Thu.) is correct.

```
Today: 2024 09 03 2
How many bottles of the juice: 47
7-digit number: 0512002

Finish all the juice on 2024/10/04 (Fri.).
```

  - Check the condition and value of the bottle_juice. → Need subtract once before entering the while loop.

```
//calculate which day finish juice
bottle_juice -= (digit_7/static_cast<int>(pow(10,7-week)))%10;

while( bottle_juice > 0 )
{
    week = (week == 7) ? 1 : week+1;
    bottle_juice -= (digit_7/static_cast<int>(pow(10,7-week)))%10;
    switch ( month )
    {
```

Lab 5.4

P113020. Lab 5.4: Monopoly Game (20%)

- Test target: loop, cin.get()
- You are asked to write a program to play "Monopoly Game".



Lab 5.4

P113020. Lab 5.4: Monopoly Game (20%)

- Input
  - The height and width of the rectangular route
  - Sequence, composed of symbols or letters, which represents the order of rewards, penalties, or nothing on corresponding grids
  - Players A and B take turns entering the number of steps to move
- Output
  - Error message "Out of the bound. Enter again."
  - Print the position and score of players A and B after each run.
  - Print "Player XX wins the game!"(XX is A or B.) or "The two players are tied!".



Lab 5.4

P113020. Lab 5.4: Monopoly Game (20%)

- Thought - 1
- **How to input the sequence?**
- Use cin.get() and store them in eight char variables.
- And use a variable to record the length of the sequence

```
//2. input the sequence
char temp_seq;
char seq1, seq2, seq3, seq4, seq5, seq6, seq7, seq8;//store the sequence
int numberOfseq = 0;
```

```
cout << "Enter Sequence: ";
while( (temp_seq = cin.get()) != '\n' )
{
    numberOfseq++;
    switch( numberOfseq )
    {
        case 1:
            seq1 = temp_seq;
            break;
        case 2:
            seq2 = temp_seq;
            break;
        case 3:
            seq3 = temp_seq;
            break;
        case 4:
            seq4 = temp_seq;
            break;
        case 5:
            seq5 = temp_seq;
            break;
        case 6:
            seq6 = temp_seq;
            break;
        case 7:
            seq7 = temp_seq;
            break;
        case 8:
            seq8 = temp_seq;
            break;
        default:
            cout << "error numberOfseq ~" << endl;
    }//end switch
}//end while
```


Lab 5.4

P113020. Lab 5.4: Monopoly Game (20%)

- Thought - 2
- **How to run the game?**
- Use while loop to control. If someone's position is bigger than or equal to the length_route exit the game.

<Pseudo code>

```
//3. Start game

//be careful that the total length of the route may exceed INT_MAX
long long int length_route = 2*(height + width - 2);
long long int pos_A = 0, pos_B = 0;
long long int score_A = 100, score_B = 100;

cout << "\nStart Game!";
while( pos_A < length_route && pos_B < length_route )
{
    cout << "\n";
    //A moves
    calculate player A position;
    if ( pos_A < length_route )
        calculate player A score;

    //B moves
    calculate player B position;
    if ( pos_B < length_route )
        calculate player B score;

    //print the result in this run
    cout << "A and B are at position " << pos_A << " and " << pos_B << ".\n";
    cout << "A and B have score " << score_A << " and " << score_B << ".\n";
}//end while
```

Lab 5.4

P113020. Lab 5.4: Monopoly Game (20%)

- Thought - 3
- **How to calculate the player's position?**
- Use while loop to do fool-proof approach. Check if the position is on the corner of the route. If yes, move back two steps.

Take Player A for example

```
int move_steps;
do{
    cout << "A move: ";
    cin >> move_steps;
    if ( move_steps < 1 || move_steps > 6 )
        cout << "Out of the bound. Enter again." << endl;
}while( move_steps < 1 || move_steps > 6 );
```

Fool-proof for moving steps.

```
pos_A += move_steps;

if ( pos_A == height-1 )
    pos_A -= 2;
else if ( pos_A == height+width-2 )
    pos_A -= 2;
else if ( pos_A == 2*height+width-3 )
    pos_A -= 2;

//stop at the ending grid
if ( pos_A > 2*(height + width - 2) )
    pos_A = 2*(height + width - 2);
```

Player lands on the corner.

Lab 5.4

P113020. Lab 5.4: Monopoly Game (20%)

- Thought - 4
- **How to calculate the player's score?**
- Find the position's corresponding symbol. And update the score.

Find the position's corresponding symbol.
→ Due to repeating the sequence, use % to find the corresponding symbol.

→ 'a' can be seen as ASCII Code. After char minus char, it will get a integer.

Take Player A for example

```
char grid_score_seq;
int i_seq = (pos_A%numberOfseq == 0)? numberOfseq : pos_A%numberOfseq;

switch( i_seq )
{
    case 1:
        grid_score_seq = seq1;
        break;
    case 2:
        grid_score_seq = seq2;
        break;
    case 3:
        grid_score_seq = seq3;
        break;
    case 4:
        grid_score_seq = seq4;
        break;
    case 5:
        grid_score_seq = seq5;
        break;
    case 6:
        grid_score_seq = seq6;
        break;
    case 7:
        grid_score_seq = seq7;
        break;
    case 8:
        grid_score_seq = seq8;
        break;
    default:
        cout << "error move ~" << endl;
}//end switch

if ( grid_score_seq == '-' )
    score_A -= score_A;
else if ( grid_score_seq >= 'a' && grid_score_seq <= 'z' )
    score_A += grid_score_seq - 'a' + 1;
else if ( grid_score_seq >= 'A' && grid_score_seq <= 'Z' )
    score_A -= grid_score_seq - 'A' + 1;
```

Evaluation

- Lab5.1(30%)
  - 5 testcases, 6% each
- Lab5.2(30%)
  - 5 testcases, 6% each
- Lab5.3(30%)
  - 5 testcases, 6% each
- Lab5.4(20%)
  - 5 testcases, 4% each

Question

- Lab 5.1
  - Int num1乘int num2還是用int暫存，之後才會assign給long long int mul，所以依然會超出可儲存範圍。

```
int num1, num2;
long long int mul = num1*num2;
```

  - pow()的值會是回傳double型態，直接印會是用科學記號印出來，需要轉換成long long int再印出。
- Lab 5.2
  - 如何印出每一個case，需要知道每一種數量各吃了幾天，按照由餅乾數量小到大的順序印出。
- Lab 5.3
  - debugger設定的路徑，需要按照你下載codeblocks的位置，做調整。

D:\0_program\CodeBlocks\MinGW\bin\gdb.exe

下載位置

Question

- Lab 5.4
  - 如果一開始用的是cin來讀進寬、高，接者是用cin.get()讀入sequence每個symbol，因為cin讀取輸入數據時，遇到換行符號就停止，所以'\n'換行符號，還留在輸入緩衝區，所以接下來遇到的第一個cin.get()，讀取到的會是留下來的'\n'換行符號，才會開始讀sequence，因此為了保持輸入讀取正確會在讀寬高和讀sequence中間多加一次cin.get()，吃掉留在緩衝區的'\n'換行符號！ (也可以用cin.ignore()解決 ref: https://junwu.nptu.edu.tw/dokuwiki/doku.php?id=cpp:strings)

```
//1. input the height and width of the rectangular route
long long int height, width;//store the height and width of the rectangular route
cout << "Enter height and width: ";
cin >> height >> width;
cin.get();//get the newline character '\n'

//2. input the sequence
char temp_seq;
char seq1, seq2, seq3, seq4, seq5, seq6, seq7, seq8;//store the sequence
int numberOfseq = 0;
cout << "Enter Sequence: ";
while( (temp_seq = cin.get()) != '\n' )
{
```






Andy, Yu-Guang Chen

Assistant Professor, Department of EE, NCU

Email: andyygchen@ee.ncu.edu.tw